7. Natural gas is a source of methane.

(a) Methane, CH_4, can be used as a fuel.

In an experiment, methane was burned to raise the temperature of 100 cm^3 of water by 27 °C.

Using the enthalpy of combustion of methane (891 $kJ\,mol^{-1}$), calculate the mass of methane, in g, burned in this experiment. **3**

(b) The equation for the combustion of methane is shown.

$$CH_4(g) \quad + \quad 2O_2(g) \quad \rightarrow \quad CO_2(g) \quad + \quad 2H_2O(g)$$

Bond enthalpies can be used to calculate a theoretical enthalpy change for this reaction.

Using bond enthalpies from the data booklet, calculate the enthalpy change, in $kJ\,mol^{-1}$, for the combustion of methane. **2**

MARKS | DO NOT WRITE IN THIS MARGIN

7. (continued)

(c) Methane reacts with steam to produce hydrogen.

$$\underset{GFM = 16\text{ g}}{CH_4(g)} + \underset{GFM = 18\text{ g}}{H_2O(g)} \rightarrow \underset{GFM = 28\text{ g}}{CO(g)} + \underset{GFM = 2\text{ g}}{3H_2(g)}$$

Calculate the atom economy for the formation of hydrogen. **2**

(d) Another naturally occurring gas is nitrogen dioxide, NO_2.

Nitrogen dioxide exists in equilibrium with dinitrogen tetroxide, N_2O_4.

$$2NO_2(g) \rightleftharpoons N_2O_4(g) \qquad \Delta H = -58\text{ kJ mol}^{-1}$$

Complete the table to show the conditions that would maximise the yield of **nitrogen dioxide**. **1**

Condition	High/Low
Temperature	
Pressure	

[Turn over

X813760119

7. (continued)

(e) (i) In the United States Space Shuttle, dinitrogen tetroxide was reacted with methylhydrazine.

$$4CH_3NHNH_2(\ell) + 5N_2O_4(\ell) \rightarrow 4CO_2(g) + 12H_2O(g) + 9N_2(g)$$

Calculate the enthalpy of this reaction, in kJ, by using the data shown below. **2**

$C(s) + 3H_2(g) + N_2(g)$	$\rightarrow$	$CH_3NHNH_2(\ell)$	$\Delta H = +54\ kJ\ mol^{-1}$
$N_2(g) + 2O_2(g)$	$\rightarrow$	$N_2O_4(\ell)$	$\Delta H = -20\ kJ\ mol^{-1}$
$C(s) + O_2(g)$	$\rightarrow$	$CO_2(g)$	$\Delta H = -394\ kJ\ mol^{-1}$
$H_2(g) + \frac{1}{2}O_2(g)$	$\rightarrow$	$H_2O(\ell)$	$\Delta H = -286\ kJ\ mol^{-1}$
$H_2O(l)$	$\rightarrow$	$H_2O(g)$	$\Delta H = +41\ kJ\ mol^{-1}$

(ii) Draw the full structural formula for methylhydrazine, CH_3NHNH_2. **1**